This guide covers the mathematics you should ideally complete before your first year or second year of university. Some topics appear more than once, and this is deliberate: seeing the same ideas presented in different styles deepens your understanding, and the extra practice makes them second nature. Whatever you pursue at university, your foundations must be strong, because everything else is built on them. If you have a better or equivalent alternative, use that. You may also download a PDF where available, or obtain the books by any other means. There is no need to waste money.

1. We begin with basic arithmetic and prepare for a course in algebra.
    - *Prealgebra* by AoPS
2. The Gelfand Correspondence program, the For the Practical Man series as well as other books for a first pass in basic mathematics.
    - *Algebra* by Gelfand and Shen and *Algebra for the Practical Man*
    - *Intermediate Algebra* by AoPS and *Introduction to Counting and Probability* by AoPS
    - *The Method of Coordinates* by Gelfand, Glagoleva, and Kirillov
    - *Functions and Graphs* by Gelfand, Glagoleva, and Shnol
    - *Trigonometry* by Gelfand and Saul and *Trigonometry* by Loney
    - *Elements of Coordinate Geometry* by Loney
    - *Geometry* by Gelfand and Alekseyevskaya and *Planimetry and Stereometry* by Kiselev
3. Next we learn how to do basic proofs in mathematics.
    - *How to Prove It* by Velleman
    - *Book of Proof* by Hammack
    - *Elementary Number Theory* by Burton
4. Next, we cover topics you will meet early in university, along with the background needed for a first pass at calculus.
    - *Discrete Mathematics* by Epp or *Introductory Discrete Mathematics* by Balakrishnan
    - *Combinatorics* by Mladenovic and *Principles and Techniques in Combinatorics* by Chen, Koh
    - *Linear Algbera and its Applications* by Strang and *About Vectors* by Hoffman
    - *Complex Numbers* by Andresscu, Andrica
    - *Precalculus* by Axler
5. We can now take a first pass at calculus and analysis.
    - *Calculus: A Complete Course* by Adams, Essex or *The Calculus with Analytic Geometry* by Louis Leithold
    - *How to Think About Analysis* by Alcock and *Understanding Analysis* by Abbott

Well a guide is incomplete without recommending problem books for the various topics mentioned here.

- *Problems in Elementary Mathematics* by Lidsky
- *A Problem Book in Algebra* by Krechmar
- *104 Trigonometry Problems* by Adreesscu, Andrica
- *Problems in Plane Geometry* by Sharygin and *Problems in Plane Geometry* by Prasolov
- *104 Number Theory Problems* by Andreescu, Andrica, Feng and *Problems in Number Theory* by Sierpinski
- *Problems in Complex Numbers* by Prasolov
- *The USSR Olympiad Problem Book* by Shklyarsky, Chentkov, Yaglov
- *Problem-Solving through Problems* by Larson
- *Problems and Theorem in Linear Algebra* by Prasolov
- *Problems in Algebra, Arithmetic and Analysis* by Prasolov
- *Problems in Calculus of One Variable* by I.A. Maron
- *Calculus with Multiple Variables* by McMullen
- *Problems in Analytic Geometry* by Kletenik
- *Problems in Mathematics* by Govorov, Dybov, Miroshin, Smibnova
- *A Problem Book in Mathematical Analysis* by Berman
- *Fifty Challenging Problems in Probability* by Frederick Mosteller